

Bankrate

BI Manager Case Study

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Part 1:

BI and Data Infrastructure

Bringing the data together in Databricks and Looker



Source tables

users
sessions
partners
applications

mortgage_clickouts
clickouts
mortgage_revenue
revenue

Tables have foreign keys for clean joins to one another. Some tables, such as sessions and clickouts, may require **custom keys** to link.

There is opportunity to **centralize revenue and clickouts data** in Databricks for cleaner modeling, rather than relying on split tables.



Upstream transforms

Create **fct_revenue** table (**mortgage_revenue** and **revenue**) and **fct_clickouts** table (**mortgage_clickouts** and **clickouts**)

- product_type will differentiate the source

Create foreign key in **clickouts** table by deriving a *session_id* from *session_start_time* and *clickout_time*

Generally, complex transformations will take place upstream, while aggregations and custom measures will take place in Looker



Looker modeling

1. Import tables into Looker as fact and dimension views

- dim_users.view.lkml
- dim_partners.view.lkml
- fct_sessions.view.lkml
- fct_applications.view.lkml
- fct_clickouts.view.lkml
- fct_revenue.view.lkml

2. Create Explores in Model for specific context, limiting to 3-5 views in each

- Revenue
- Clickout Funnel
- Applications
- Sessions

What users see and what they don't



Visible to users

- Most dimensions and measures should be accessible, but this should be intentional to limit sprawl and maintain security
- IDs necessary for granular detail (e.g. application_id)
- FICO and income shown as grouped buckets
- Clear classifiers (e.g. "Approved / Declined" instead of 1/0)



Not visible to users

- IDs and keys that are only necessary for linking data
- IDs that are superseded by a name (e.g. partner_name)
- PII and sensitive data, such as income and FICO (permissioned access only)
- Unclear boolean or 1/0 measures

Leadership KPI Dashboard

Partner ▾

Channel ▾

FICO Score Group ▾

Total Revenue (TTM)

\$2.64M

YoY +8.2%
QoQ +3.1%
MoM +1.4%

Approvals (TTM)

10,976

YoY +6.5%
QoQ +2.4%
MoM +0.9%

Approval Rate (TTM)

11.0%

YoY +1.1%
QoQ +0.5%
MoM +0.2%

Revenue / Application (TTM)

\$26.42

YoY +1.6%
QoQ +0.7%
MoM +0.4%

Acquisition Funnel

Sessions



Clickouts



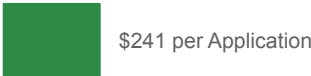
Applications



Approvals



Revenue



Total

1,200,000

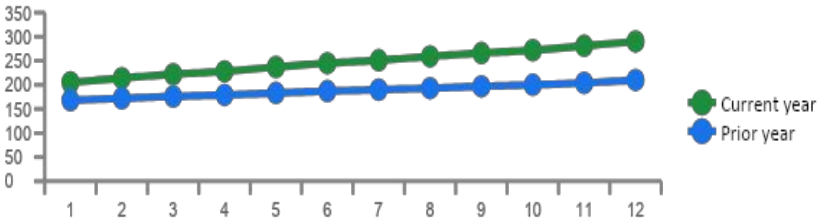
420,000

100,000

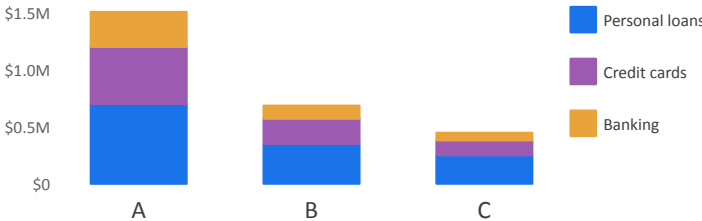
10,976

\$2.64M

Revenue Trend (YoY)



Revenue by Partner & Product



Core Aggregate Views

Executive Leadership

Focuses on overall performance and business health to determine go-forward strategy

- Revenue Fact Explore
- Acquisition Funnel Explore

Growth & Marketing

Focuses on acquiring customers, increasing applications and revenue, and marketing campaigns

- Acquisition Funnel Explore
- Application Explore
- Session Explore
- Marketing Campaign Explore

Partner/Revenue Management

Focuses on partner performance and revenue streams to inform relationships and deals and prioritize customer flow

- Revenue Fact Explore
- Partner Explore

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Part 2:

Partner Matching Analysis

Partner funnel overview

Hypothesis: The current customer-to-partner matching process is suboptimal. There is an opportunity to maximize revenue by better pairing customers with partners and increasing approvals, especially given customer data, approval rates, and payouts per loan.

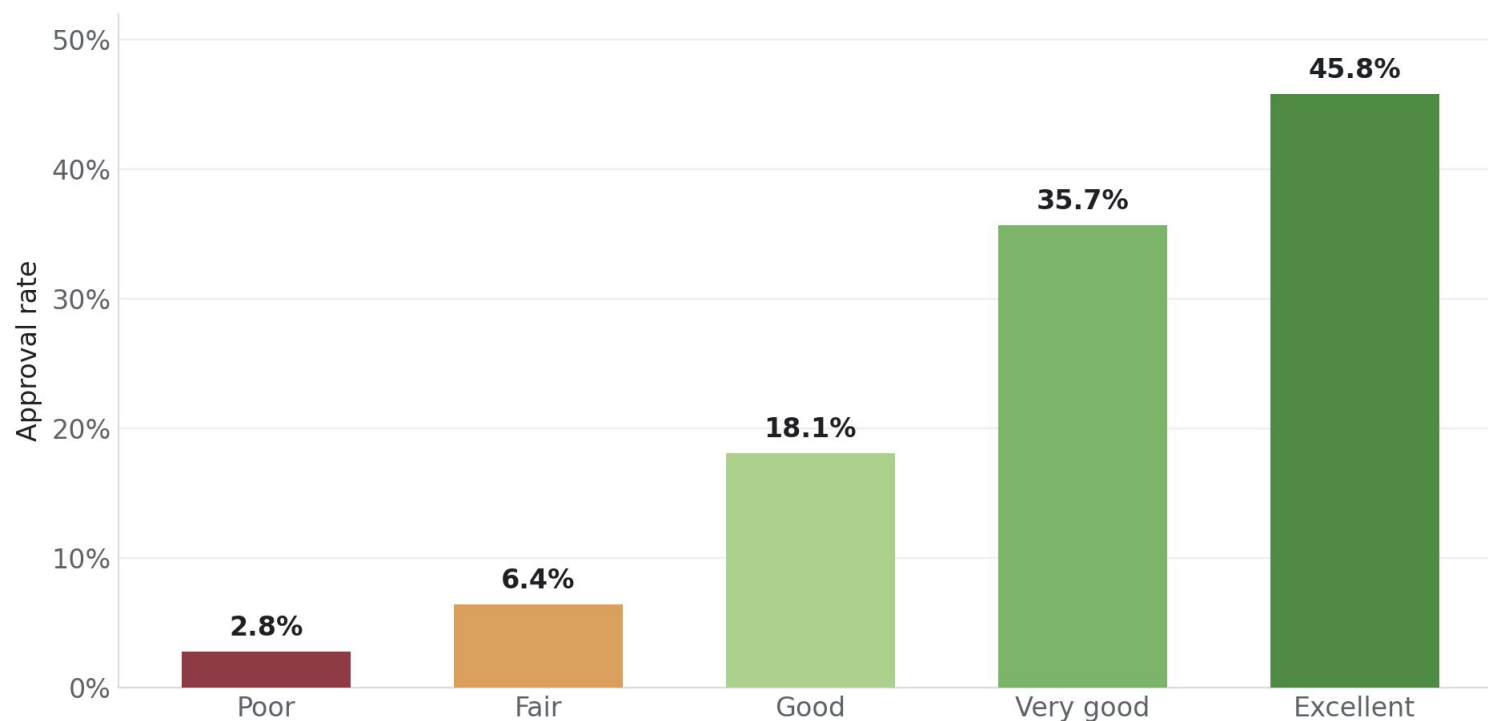
Partner	Applications	Payout per Approval	Approval Rate	Revenue	Revenue per Application
A	55,000 (55%)	\$250	11.0%	\$1,507,750	\$27.41
B	27,500 (27%)	\$350	7.1%	\$686,000	\$24.95
C	17,500 (18%)	\$150	17.1%	\$447,750	\$25.59
Total	100,000 (100%)	—	11.0%	\$2,641,500	\$26.42

Partner A has the **most applications** and **highest RPA** compared to other partners, but sits in the middle in terms of payout and approval rate. **Partner B** has roughly a quarter of all applications and the **highest payout**, but the **lowest approval rate and lowest RPA**. **Partner C** has the fewest applications and **lowest payout**, but the **highest approval rate**.

In order to determine how to optimally match customers to partners, we need to better understand the customer data.

What determines approvability?

Approval rate by FICO score



Housing Ratio

Ideal ($\leq 28\%$) represented in 48% of applications with a 15% approval rate

Prior Bankruptcy or Foreclosure

97% of applications do not have prior bankruptcy or foreclosure

Employment

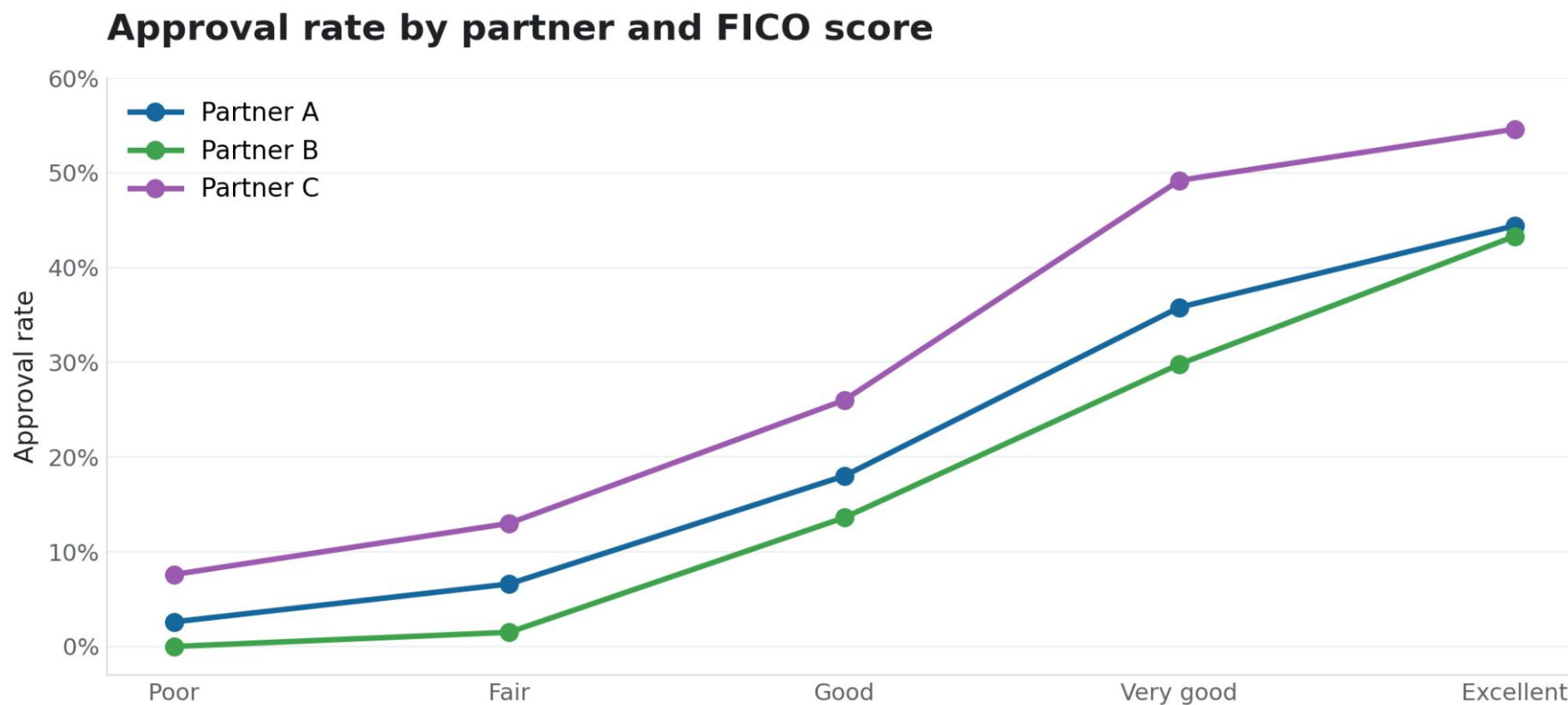
77% of customers are FTE with 12% approval rate

Partners and FICO score distribution

FICO scores are distributed similarly across partners



Approval rates differ across partners and FICO



Recommendations

- Match **lower FICO scores** with **Partner C**, which approves **2x** as often as Partner A, and avoid matching with Partner B.
- Match **higher FICO scores** with **Partner B** to maximize expected revenue (\$350) from more valuable applications.
- **Partner A** funnel, which receives 55% of applications, provides **meaningful volume to redistribute**.
- Test and **validate updated routing** against a subset of customers and refine model before full deployment.